A Low-p Zero-Boundary Result for Sobolev-to-BV Extension Domains

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Source Problem

Rajala and Zhu ask whether the boundary of a $(W^{1,p}, BV)$-extension domain must have Lebesgue measure zero in the remaining range $1 < p \leq (n-1)/(n-2)$ [1, Question 1.3]. The question appears on PDF page 3 of arXiv:2205.10801 and in the parsed source at `main.tex:308-312`.

Theorem 1.1. *A domain $\Omega \subset \mathbb{R}^n$ is a (BV, BV)-extension domain if and only if it is a $\left(W^{1,1}, BV\right)$-extension domain.*

Since, $W^{1,1}(\Omega)$ is a proper subspace of $BV(\Omega)$ with $\|u\|_{W^{1,1}(\Omega)} = \|u\|_{BV(\Omega)}$ for every $u \in W^{1,1}(\Omega)$, (BV, BV)-extension property implies $(W^{1,1}, BV)$-extension property immediately. The other direction from $(W^{1,1}, BV)$-extension property to (BV, BV)-extension property is not as straightforward, as $W^{1,1}(\Omega)$ is only a proper subspace of $BV(\Omega)$. The essential tool here is the Whitney smoothing operator constructed by García-Bravo and the first named author in [GBR21]. This Whitney smoothing operator maps every function in $BV(\Omega)$ to a function in $W^{1,1}(\Omega)$ with the same trace on $\partial\Omega$, so that the norm of the image in $W^{1,1}(\Omega)$ is uniformly controlled from above by the norm of the corresponding preimage in $BV(\Omega)$.

With an extra assumption that Ω is q-fat at almost every point on the boundary $\partial\Omega$, in [KUZ21] it was shown that the boundary of a $(W^{1,p}, W^{1,q})$-extension domain is of volume zero when $1 \le q < p < \infty$. The essential point there was that the q-fatness of the domain on the boundary guarantees the continuity of a $W^{1,q}$-function on the boundary. Maybe a bit surprisingly, the assumption that the domain is 1-fat at almost every point on the boundary also guarantees that the boundary of a $(W^{1,p}, BV)$-extension domain is of volume zero. In particular, every planar domain is 1-fat at every point of the boundary. Hence, we have the following theorem.

Theorem 1.2. *Let $\Omega \subset \mathbb{R}^n$ be a $(W^{1,p}, BV)$-extension domain for $1 \le p < \infty$, which is 1-fat at almost every $x \in \partial\Omega$. Then $|\partial\Omega| = 0$. In particular, for every planar $(W^{1,p}, BV)$-extension domain Ω with $1 \le p < \infty$, we have $|\partial\Omega| = 0$.*

In light of the results and example given in [KUZ21], the most interesting open question is what happens in the range $1 < p \le (n-1)/(n-2)$ of exponents. For this range, we do not know whether the boundary of a $(W^{1,p}, BV)$-extension domain must be of volume zero. If a counterexample exists in this range, it might be easier to construct it in the $(W^{1,p}, BV)$-case rather than the $(W^{1,p}, W^{1,1})$-case. Hence we leave it as a question here.

Question 1.3. *For $1 < p \le (n-1)/(n-2)$, is the boundary of a $(W^{1,p}, BV)$-extension domain of volume zero?*

2. Preliminaries

For a locally integrable function $u \in L^1_{\mathrm{loc}}(\Omega)$ and a measurable subset $A \subset \Omega$ with $0 < |A| < \infty$, we define

$$u_A := \fint_E u(y)\, dy = \frac{1}{|A|} \int_A u(y)\, dy.$$

Definition 2.1. *Let $\Omega \subset \mathbb{R}^n$ be a domain. For every $1 \le p \le \infty$, we define the Sobolev space $W^{1,p}(\Omega)$ to be*

$$W^{1,p}(\Omega) := \{u \in L^p(\Omega) : \nabla u \in L^p(\Omega; \mathbb{R}^n)\},$$

where ∇u denotes the distributional gradient of u. It is equipped with the nonhomogeneous norm

$$\|u\|_{W^{1,p}(\Omega)} = \|u\|_{L^p(\Omega)} + \|\nabla u\|_{L^p(\Omega)}.$$

Now, let us give the definition of functions of bounded variation.

Result

This packet proves the zero-boundary conclusion in a strict low-p subrange of the open interval, without the 1-fatness assumption used in the source paper.

Theorem 1. *Let $n \geq 2$, let $\Omega \subset \mathbb{R}^n$ be a bounded $(W^{1,p}, BV)$-extension domain, and assume*

$$1 < p < 1^* := \frac{n}{n-1}.$$

Then $|\partial\Omega| = 0$.

For $n \geq 3$, this covers a genuine subrange of Rajala–Zhu's Question 1.3, because

$$\frac{n}{n-1} < \frac{n-1}{n-2}.$$

For $n = 2$, Rajala–Zhu already prove the stronger planar statement for every $1 \leq p < \infty$.

Definitions and Imported Facts

A domain $\Omega \subset \mathbb{R}^n$ is a $(W^{1,p}, BV)$-extension domain if there is an operator $E : W^{1,p}(\Omega) \to BV(\mathbb{R}^n)$ such that $E(u) = u$ on Ω and

$$\|E(u)\|_{BV(\mathbb{R}^n)} \leq C_E \|u\|_{W^{1,p}(\Omega)}$$

for all $u \in W^{1,p}(\Omega)$. Linearity is not assumed.

We use three standard ingredients. First, the global Sobolev embedding for functions of bounded variation gives

$$\|w\|_{L^{1^*}(\mathbb{R}^n)} \leq C_n \|w\|_{BV(\mathbb{R}^n)}.$$

Second, the local BV Sobolev-Poincare inequality on balls gives

$$\inf_{a \in \mathbb{R}} \|w - a\|_{L^{1^*}(B)} \leq C_n \|Dw\|(B) \qquad (w \in BV(B)).$$

Third, Rajala–Zhu define a bounded quasiadditive set function Φ for $(W^{1,p}, BV)$-extension domains [1, Lemmas 3.2–3.4]. For open $U \subset \mathbb{R}^n$, set

$$W_0^p(U, \Omega) = \{u \in W^{1,p}(\Omega) \cap C(\Omega) : u = 0 \text{ on } \Omega \setminus U\},$$

$$\Gamma(u) = \inf\{\|Dv\|(U) : v \in BV(\mathbb{R}^n),\ v|_\Omega = u\},$$

and

$$\Phi(U) = \sup_{u \in W_0^p(U,\Omega)} \left(\frac{\Gamma(u)}{\|u\|_{W^{1,p}(U \cap \Omega)}} \right)^k, \qquad \frac{1}{k} = 1 - \frac{1}{p}.$$

Then $\overline{D\Phi}(x) < \infty$ for almost every $x \in \mathbb{R}^n$, where

$$\overline{D\Phi}(x) = \limsup_{r \downarrow 0} \frac{\Phi(B(x,r))}{|B(x,r)|}.$$

Moreover, for every ball $B(x,r)$ and every $u \in W_0^p(B(x,r), \Omega)$ there is $v \in BV(B(x,r))$ with $v = u$ on $B(x,r) \cap \Omega$ and

$$\|Dv\|(B(x,r)) \leq 2\Phi(B(x,r))^{1/k} \|u\|_{W^{1,p}(B(x,r) \cap \Omega)}.$$

Proof Intuition

Koskela and Mishra prove that a bounded $(W^{1,p}, W^{1,q})$-extension domain has zero boundary measure when $1 \le q < p < q^* = qn/(n-q)$ [2]. Their argument has two parts: a lower density bound obtained from the global Sobolev embedding, and an upper decay iteration at boundary density points obtained from a local set-function extension estimate.

For BV, the role of q^* is played by the critical BV exponent $1^* = n/(n-1)$. The Rajala–Zhu set function gives exactly the local variation estimate needed for the upper decay half of the argument. Thus the Koskela–Mishra iteration adapts when $p < 1^*$. The strict inequality is essential in this method: it is what makes small boundary-density volume absorb the right side of the BV Sobolev-Poincare estimate.

Proof

Put

$$m_x(r) = |B(x,r) \cap \Omega|.$$

Lemma 1 (Lower content bound)**.** *Assume* $1 < p < 1^*$*. There are constants* $\delta > 0$ *and*

$$s = \left(\frac{1}{p} - \frac{1}{1^*}\right)^{-1} \ge n$$

such that

$$m_x(r) \ge \delta r^s$$

for every $x \in \partial\Omega$ *and all sufficiently small* $r > 0$*.*

Proof. The proof is the BV analogue of the first density lemma of Koskela–Mishra [2, Lemma 3.1]. Let $x \in \partial\Omega$ and fix a small $r > 0$. Choose radii

$$r = r_0 > r_1 > r_2 > \cdots \downarrow 0$$

so that

$$m_x(r_j) = 2^{-j} m_x(r).$$

If necessary, one chooses continuity radii for m_x and then passes to the limit; this does not change the estimate.

For $j \ge 1$, let u_j be the Lipschitz cutoff on Ω which equals 1 on $B(x, r_j) \cap \Omega$, vanishes outside $B(x, r_{j-1}) \cap \Omega$, and has Lipschitz constant at most $(r_{j-1} - r_j)^{-1}$. Since r is small,

$$\|u_j\|_{W^{1,p}(\Omega)} \le C(r_{j-1} - r_j)^{-1} m_x(r_{j-1})^{1/p}.$$

Let Eu_j be a global BV extension. By the BV Sobolev embedding and the boundedness of the extension operator,

$$m_x(r_j)^{1/1^*} \le \|Eu_j\|_{L^{1^*}(\mathbb{R}^n)} \le C\|Eu_j\|_{BV(\mathbb{R}^n)} \le C(r_{j-1} - r_j)^{-1} m_x(r_{j-1})^{1/p}.$$

Using $m_x(r_j) = 2^{-j} m_x(r)$, this gives

$$r_{j-1} - r_j \le C\, m_x(r)^{1/p - 1/1^*} 2^{-j(1/p - 1/1^*)}.$$

Since $p < 1^*$, the geometric series converges, and summing in j yields

$$r \le C\, m_x(r)^{1/p - 1/1^*}.$$

Equivalently $m_x(r) \ge \delta r^s$, with $s = (1/p - 1/1^*)^{-1}$. $\square$

For $x \in \partial\Omega$, define the annulus

$$A_{r_1,r_2,\Omega} = \{y \in \Omega : r_1 < |y - x| < r_2\}.$$

Lemma 2 (Small-density annular estimate)**.** *Let $x \in \partial\Omega$ satisfy*

$$\limsup_{r\downarrow 0} \frac{m_x(r)}{r^n} = 0 \qquad \textit{and} \qquad \overline{D\Phi}(x) < \infty.$$

Then for every $c > 0$ there is $r_{x,c} > 0$ such that, whenever $0 < r < r_{x,c}$,

$$\min\{|A_{r,2r,\Omega}|,\ m_x(r/2)\} \le c\, m_x(r).$$

Proof. Let $a = 1^* = n/(n-1)$. Since $\overline{D\Phi}(x) < \infty$, for all small r,

$$\Phi(B(x, 2r))^{1/k} \le C_x r^{n/k} = C_x r^{\,n-n/p}.$$

Let $u \in W_0^p(B(x, 2r), \Omega)$ be the radial cutoff satisfying

$$u = 1 \text{ on } B(x, r/2) \cap \Omega, \qquad u = 0 \text{ on } \Omega \setminus B(x, r), \qquad |\nabla u| \le 2/r.$$

Rajala–Zhu's local extension lemma gives $v \in BV(B(x, 2r))$, equal to u on $B(x, 2r) \cap \Omega$, with

$$\|Dv\|(B(x, 2r)) \le C_x r^{\,n-n/p} \left(m_x(r)^{1/p} + r^{-1}|A_{r/2,r,\Omega}|^{1/p}\right).$$

The function v equals 1 on $B(x, r/2) \cap \Omega$ and 0 on $A_{r,2r,\Omega}$. Applying the BV Sobolev-Poincare inequality to $v - v_{B(x,2r)}$ gives

$$\begin{aligned}\min\{|A_{r,2r,\Omega}|^{1/a}, m_x(r/2)^{1/a}\} &\le C\|Dv\|(B(x, 2r)) \\ &\le C_x \left(r^{\,n-n/p} m_x(r)^{1/p} + r^{\,n-n/p-1}|A_{r/2,r,\Omega}|^{1/p}\right).\end{aligned}$$

Set $Y = \min\{|A_{r,2r,\Omega}|, m_x(r/2)\}$. Since $|A_{r/2,r,\Omega}| \le m_x(r)$, raising the last inequality to the power a gives

$$Y \le C_x \left(r^{(n-n/p)a} m_x(r)^{a/p} + r^{(n-n/p-1)a} m_x(r)^{a/p}\right).$$

The density assumption says that for every $\eta > 0$, after decreasing $r_{x,\eta}$ we have $m_x(r) \le \eta r^n$. Because $p < a$,

$$r^{(n-n/p)a} m_x(r)^{a/p} \le C\eta^{a/p-1} m_x(r)$$

for $r < 1$, and the critical exponent identity $a = n/(n-1)$ gives

$$r^{(n-n/p-1)a} m_x(r)^{a/p} \le C\eta^{a/p-1} m_x(r).$$

Choosing η so small that $C_x\eta^{a/p-1} < c$ proves the claim. □

Lemma 3 (Annular iteration)**.** *Under the hypotheses of Lemma 2, for every $0 < c < 1/2$ there is $r_{x,c} > 0$ such that*

$$m_x(r/2) \le c\, m_x(r)$$

whenever $0 < r < r_{x,c}$.

Proof. This is the purely measure-theoretic annular iteration of Koskela–Mishra [2, Lemma 3.3], applied after Lemma 2. We recall the short argument. If the minimum in Lemma 2 is $m_x(r/2)$, there is nothing to prove. Otherwise

$$|A_{r,2r,\Omega}| \le c\, m_x(r).$$

Repeating Lemma 2 at radii $2r, 4r, \ldots$, one inductively forces the minimum to be the corresponding annular term; if at some step it were the inner ball term, the inequality

$$m_x(2^{j-1}r) \le c\big(m_x(2^{j-1}r) + |A_{2^{j-1}r,2^jr,\Omega}|\big)$$

together with the previous annular bounds would contradict $c < 1/2$. Hence

$$|A_{2^{j-1}r,2^jr,\Omega}| \le (1+c)^j m_x(r)$$

for all j with 2^jr below a fixed small radius R.

Choose j with $R/2 \le 2^{j-1}r \le R$. Since Ω is connected and $x \in \partial\Omega$, after taking $R < \mathrm{diam}(\Omega)/6$ the annuli $A_{s/2,s,\Omega}$, $R \le s \le 2R$, all have positive measure; by continuity in s their measures have a positive lower bound σ. On the other hand, the displayed upper bound and $m_x(r) \le C_n r^n$ imply

$$\sigma \le (1+c)^j C_n r^n \le C_N R\, r^{n-1},$$

which is impossible for sufficiently small r. Thus the alternative in which the minimum is the annulus cannot occur at small scales, and the desired decay $m_x(r/2) \le c\, m_x(r)$ follows. □

We now prove Theorem 1. Suppose, for contradiction, that $|\partial\Omega| > 0$. By the Lebesgue density theorem and the almost-everywhere finiteness of $\overline{D\Phi}$, there is

$$x \in \partial\Omega$$

such that x is a Lebesgue density point of $\partial\Omega$ and $\overline{D\Phi}(x) < \infty$. At such a point,

$$\frac{m_x(r)}{r^n} \to 0.$$

Let $s = (1/p - 1/1^*)^{-1}$ be the exponent from Lemma 1, and choose $c < 2^{-s}$. By Lemma 3, for all small r,

$$m_x(r/2) \le c\, m_x(r).$$

Iterating gives

$$m_x(2^{-j}r) \le c^j m_x(r).$$

The lower content bound gives, for all large j,

$$\delta(2^{-j}r)^s \le m_x(2^{-j}r) \le c^j m_x(r).$$

Equivalently,

$$\frac{\delta r^s}{m_x(r)} \le (2^s c)^j.$$

The left side is fixed and positive, while $2^s c < 1$, so the right side tends to 0, a contradiction. Therefore $|\partial\Omega| = 0$.

Verification Notes

The proof uses the Rajala–Zhu local BV extension lemma exactly in the form stated in their Lemma 3.4. The only adaptation point is replacing the $W^{1,q}$ Sobolev-Poincare step in Koskela–Mishra by the critical BV Sobolev-Poincare inequality. The exponent arithmetic is:

$$(n - n/p - 1)1^* + n(1^*/p - 1) = 0,$$

so the annular term is absorbed precisely because $1^*/p - 1 > 0$, i.e. $p < 1^*$.

Limitations and Novelty Check

This is not a full answer to Rajala–Zhu Question 1.3. It leaves open

$$\frac{n}{n-1} \leq p \leq \frac{n-1}{n-2} \qquad (n \geq 3).$$

The endpoint $p = 1^*$ is not reached by this route because the small-density absorption exponent $1^*/p - 1$ becomes zero.

The run indexes were searched for arXiv:2205.10801 and for close phrases involving $(W^{1,p}, BV)$-extension domains, zero-volume boundary, Sobolev to BV, and Question 1.3. A web search on 2026-07-18 found the source paper [1], the 2025 Koskela–Mishra theorem for $(W^{1,p}, W^{1,q})$-extension domains [2], and the earlier Koskela–Ukhlov–Zhu examples/theorem [3], but no exact statement of the $(W^{1,p}, BV)$ low-p theorem above. Since the proof is a close adaptation of [2], the novelty claim should be read conservatively: it is plausibly new as an explicit BV corollary, or at minimum a useful recorded implication not currently indexed in the run.

Human Review Recommendation

Review as a likely valid partial theorem. The most important checks are the local BV Sobolev-Poincare min-estimate in Lemma 2, the annular iteration import from Koskela–Mishra, and the endpoint obstruction at $p = n/(n-1)$. If a later literature source already states this exact $(W^{1,p}, BV)$ subcase, reclassify the packet as a literature-implied answer.

References

[1] T. Rajala and Z. Zhu, *Zero volume boundary for extension domains from Sobolev to BV*, arXiv:2205.10801; Revista Matematica Complutense, published online 2024, DOI 10.1007/s13163-024-00485-6.

[2] P. Koskela and R. Mishra, *The volume of the boundary of a Sobolev (p, q)-extension domain II*, arXiv:2409.01170; Calculus of Variations and Partial Differential Equations **64** (2025), Article 84, DOI 10.1007/s00526-025-02934-6.

[3] P. Koskela, A. Ukhlov, and Z. Zhu, *The volume of the boundary of a Sobolev (p, q)-extension domain*, Journal of Functional Analysis **283** (2022), Article 109703, DOI 10.1016/j.jfa.2022.109703.